No.

Date:

Latihan 2

Tentukan deret fourier fungsi berikut!

1. $f(t)\begin{cases} 0 & -5 < t < 0 \\ 1 & 0 < t < 5 \end{cases}$ dgn $f(t) = f(t+10)$

$T = 10, \quad \omega = \frac{2\pi}{T} = \frac{2\pi}{10} = \frac{\pi}{5}$

$$a_0 = \frac{1}{T}\int_0^T f(t)\,dt = \frac{1}{10}\left(\int_{-5}^{0} 0\,dt + \int_0^5 1\,dt\right) = \frac{1}{10}\left[0 + (t)\Big|_0^5\right]$$

$$= \frac{1}{10}(5) = \frac{1}{2}$$

$$a_n = \frac{2}{T}\int_0^T f(t)\cos n\omega t\,dt = \frac{1}{5}\left(\int_{-5}^{0} 0\cos n\frac{\pi}{5}t\,dt + \int_0^5 \cos n\frac{\pi}{5}t\,dt\right)$$

$$= \frac{1}{5}\left(0 + \left(5\,\frac{\sin\frac{\pi}{5}nt}{\pi n}\right)_0^5\right) = \frac{1}{5}\left(5\,\frac{\sin \pi n}{\pi n} - 0\right)$$

$$= \frac{\sin \pi n}{\pi n}$$

$a_n = 0$ untuk n integer

$$b_n = \frac{2}{T}\int_0^T f(t)\sin n\omega t\,dt = \frac{1}{5}\left(\int_{-5}^{0} 0\sin\frac{\pi}{5}nt\,dt + \int_0^5 \sin\frac{\pi}{5}nt\,dt\right)$$

$$= \frac{1}{5}\left(0 + \left(-\frac{5\cos\frac{\pi}{5}nt}{\pi n}\right)_0^5\right) = \frac{1}{5}\left(-\frac{5\cos \pi n}{\pi n} - \left(-\frac{5}{\pi n}\right)\right)$$

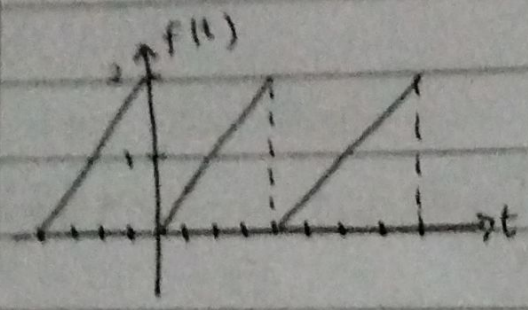
$$= \frac{1}{5}\left(\frac{-5\cos \pi n + 5}{\pi n}\right) = \frac{-5}{5}\left(\frac{\cos \pi n - 1}{\pi n}\right) = -\frac{\cos \pi n - 1}{\pi n}$$

$$b_n \begin{cases} 0 & \text{untuk } n \text{ integer genap} \\ \frac{2}{n\pi} & \text{untuk } n \text{ integer ganjil} \end{cases}$$

$f(t) = a_0 + a_1 \cos \omega t + \dots + a_n \cos n\omega t + \dots + b_1 \sin \omega t + \dots + b_n \sin n\omega t$

$f(t) = \frac{1}{2} + \frac{2}{\pi} \sin \frac{\pi}{5} t + \frac{2}{3\pi} \sin \frac{3\pi}{5} t + \frac{2}{5\pi} \sin \frac{5\pi}{5} t + \frac{2}{7\pi} \sin \frac{7\pi}{5}$

10. Diketahui $f(t)$ seperti ditunjukkan pada gambar berikut!



$f(t) = \frac{x}{2}$ $0 < t < 4$ dgn $f(t) = f(t+8)$

$\frac{x}{2} - 2$ $4 < t < 8$

$f(t) = \frac{t}{2}$ $0 < t < 4$ dgn $f(t) = f(t+4)$

$T = 4, \quad \omega = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{\pi}{2}$

$a_0 = \frac{1}{4} \int_0^4 \frac{t}{2} dt = \frac{1}{8} \left(\frac{1}{2} t^2 \right) \Big|_0^4 = 1$

$a_n = \frac{2}{4} \int_0^4 \frac{t}{2} \cos \frac{\pi}{2} nt \, dt$

$= \frac{2}{8} \int_0^4 t \cos \frac{\pi}{2} nt \, dt$

$= \frac{1}{4} \left(\frac{2t \sin \left(\frac{\pi}{2} nt \right)}{\pi n} + \frac{4 \cos \left(\frac{\pi}{2} nt \right)}{n^2 \pi^2} \right) \Big|_0^4$

$= \frac{1}{4} \left(\frac{4 (2\pi n \sin (2\pi n) + \cos (2\pi n) - 1)}{\pi^2 n^2} \right)$

$= \frac{2\pi n \sin 2\pi n + \cos 2\pi n - 1}{\pi^2 n^2}$

$a_n = 0$ untuk n integer

No.

Date:

$$b_n = \frac{2}{4} \int_0^4 \frac{t}{2} \sin \frac{\pi}{2} nt$$

$$= \frac{1}{4} \int_0^4 t \sin \frac{\pi}{2} nt$$

$$= \frac{1}{4} \left(- \frac{2t \cos\left(\frac{\pi}{2} nt\right)}{\pi n} + \frac{4 \sin\left(\frac{\pi}{2} nt\right)}{n^2 \pi^2} \right) \Bigg|_0^4$$

$$= \frac{1}{4} \left(\frac{-4(2\pi n \cos 2\pi n - \sin 2\pi n)}{n^2 \pi^2} \right)$$

$$= - \frac{2\pi n \cos 2\pi n - \sin 2\pi n}{n^2 \pi^2}$$

$b_n = - \frac{2}{n\pi}$ untuk n integer

$$f(t) = 1 - \frac{2}{\pi} \sin \frac{\pi}{2} t - \frac{2}{2\pi} \sin \frac{2\pi}{2} t - \frac{2}{3\pi} \sin \frac{3\pi}{2} t$$